

CSA0711

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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I P. Sashan (C19242537) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sashan

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	2	2	2	2	2
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1/2	1/2	1/2	1/2	1/2

28/30

28/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

Smart Campus Network Design Using IPv4 Subnetting

Understanding of the Scenario

A newly established smart campus is deploying a secure and efficient network infrastructure to support academic and administrative activities. The campus has been allocated the IPv4 network 192.168.100.0/24, which contains 256 IP addresses ranging from 192.168.100.0 to 192.168.100.255. To accommodate different departments based on their expected number of devices

- School of Computing $\rightarrow$ 125
- School of Electronics $\rightarrow$ 126
- School of Mechanical Engineering $\rightarrow$ 127

As the network engineer, the objective is to divide the given network into suitable subnets and determine the network addresses, broadcast addresses.

- 1 Allocating sufficient IP addresses for each department according to the assigned subnet mask.
- 1 calculating the usable host capacity for each department.
- 1 preserving the remaining IP address space for future campus expansion, additional department.

Alloca School of Electronics C126

192.168.100.128

A126 subnet contains:

$$2^{(32-26)} = 64$$

addresses

Usable hosts:

$$64 - 2 = 62$$

Network Address

192.168.100.128

Broadcast Address:

192.168.100.191

Valid Host Range

192.168.100.129

to

192.168.100.190

Alloca School of mechanical engineering

A127 Subnet contains:

$$2^{(32-27)} = 32$$

addresses.

Usable hosts:

$$32 - 2 = 30$$

192.168.100.192

192.168.100.193

to

192.168.100.222,

Application of networking Concept

Original Network

10.10.0.0/16

Subnet mask:

255.255.0.0

Total IP Addresses:

$$2^{(32-16)} = 65,536$$

Available Address Range:

10.10.0.0 - 10.10.255.255

Allocate Software Development Ctr

Any subnet contains:

$$2^{(32-24)} = 256$$

Addresses:

Usable Host Addresses:

$$256 - 2 = 254$$

Network Address:

10.10.0.0

Broadcast Address:

10.10.0.255

Valid Host Range:

10.10.0.1

10.10.0.254

Smart Hospital IPv6 Network Migration

Understanding of the scenario

A smart hospital is upgrading its network infrastructure from IPv4 to IPv6 to support thousands of interconnected medical devices, computers, IoT sensors, patient monitoring systems, and smart healthcare equipment.

2001:0db8:abcd:0000:0000:0000:0000/64

- > Convert the given IPv6 address into its compressed notation.
- > Expand the compressed address back into its full hexadecimal notation.
- > Identify the network prefix using the /64 prefix length.
- > Calculate the total number of host addresses supported within the subnet.
- > Determine the total number of possible host addresses available within the network.

Identification of Key Issues

- > converting the IPv6 address into its shortened valid notation without changing its value.
- > restoring the compressed address back to its original full hexadecimal format.

2
128
0000
filling all missing blocks and adding leading zeros.

0:0db1:abcd:0000:0000:0000:0000

4. Determine the network prefix

- IPv6 / 128 prefix means:
 - The first 64 bits represent the network portion
 - The last 64 bits represent the Interface ID portion

The first four hexadecimal blocks are
2001:0db1:abcd:0000

Then for the network prefix is

2001:0db1:abcd:0000::/64

or in compressed notation!

2001:db1:abcd::/64

The remaining four blocks:

0000:0000:0000:0000

Host portion!

$$128 - 64 = 64 \text{ bits}$$

$$2^{64}$$

$$2^{64} = 18,446,744,073,709,551,616$$

Application of Routing Concepts

Given Routing Table

Network Address	Prefix length
192.168.1.0/24	124
192.168.2.0/24	124
192.168.3.0/24	124

$$2^{(32-24)} = 2^8$$

Usable host addresses:

$$256 - 2 = 254$$

The valid host range for 192.168.2.0/24 is

192.168.2.1 - 192.168.2.254

Broadcast Address:

192.168.2.255

Step 2: Determine the Destination Subnet

Destination IP Address:

192.168.2.45

Subnet Mask:

255.255.255.0 (/24)

The first three octets are:

192.168.2

Office Network Design Using a Layer 3 Router

Understanding of the Scenario

A company has established two separate office departments that are connected through a layer 3 router to enable communication between different local area network (LANs).

→ Administration LAN → 192.168.10.0/24

→ Finance LAN → 192.168.10.64/24

The router interfaces are configured with the following gateway IP addresses.

- Administration Gateway → 192.168.10.1

- Finance Gateway → 192.168.10.65

→ Determine the network address and broadcast address for each LAN.

→ Identify the valid host IP address range for both LANs.

→ Suggest suitable IP addresses for the computers in each LAN.

→ Identify the default gateway that host in each LAN

should use for communication.

Good Can Address

192.168.10.63

Valid Host Range

192.168.10.1

to

192.168.10.62

The router interface is configured with:

192.168.10.1

Therefore, this IP serves as the default gateway for all host in the Administration.

Step 2: Analyze the Financial AN

192.168.10.64/26

Subnet mask:

255.255.255.192

Again each subnet contains:

$$2^6 = 64$$

Usable Hosts:

$$64 - 2 = 62$$

Network Address

192.168.10.64

valid Host Range

192.168.10.0

to

192.168.10.255

The route Interface is configured with:

192.168.10.65

Assign IP Addresses to Computers

Administration LAN

Computer

IP Address

Computer 1

192.168.10.2

Computer 2

192.168.10.3

Computer 3

192.168.10.4

Configure the Default Gateway

Every computer must use the route interface within its subnet as the

Administration LAN

Default Gateway.

192.168.10.1

Finance LAN

Default Gateway

192.168.10.65